

Tariq Khalaf

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Education

University of North Carolina at Charlotte

Expected: Dec 2028

BS in Electrical and Computer Engineering

Skills

Design & Simulation: KiCad, Fusion 360, AutoCAD, SolidWorks, PCB Design, Circuit Diagrams, VS Code, Dev-C++, Meshmixer, 3D Printing

Programming: Python, C++, Arduino, Embedded C, UART, PWM, I2C/SPI basics

Engineering Skills: Soldering, PCB bring-up/debugging, buck regulators, USB-C power/programming, motor control, circuit prototyping

Experience

Volunteer Tutoring - Mathematics

Sep 2023 - Sep 2024

- Tutored students one-on-one in Algebra and Trigonometry, adapting explanations to individual learning styles.
- Strengthened foundational concepts through practice problems, step-by-step review, and test-prep support.

Volunteer Helper - Introduction to Engineering

Jan 2026 - July 2026

- Assisted students during labs and 15+ projects by explaining engineering concepts and troubleshooting builds.
- Supported hands-on activities involving Arduino, circuits, sensors, mechanical prototypes, and problem-solving workflows.

Projects

Electronic Speed Controller Project - Academic Project

Dec 2025 - Jan 2026

- Designed a 24 V DC motor ESC supporting up to 5 A continuous current, emphasizing safe power delivery and reliability.
- Created multi-layer PCB layouts in KiCad with ground planes, power pours, via stitching, and high-current trace planning.
- Selected parts using datasheets and supplier availability to balance performance, cost, footprint, and manufacturability.

Embedded MCU Power & Programming Board - Personal Project

Jun 2026 - Jul 2026

- Designed a custom MCU board that accepts 24-36 V input and steps it down to 3.3 V using an onboard buck regulator.
- Integrated USB-C for board power/programming with required power, programming, and protection support circuitry.
- Planned PCB layout around regulator placement, decoupling, grounding, and clean 3.3 V delivery for embedded hardware.

Remote-Control Robot - Personal Project

Jan 2026 - Feb 2026

- Combined RC car and robot concepts into a mobile platform using microcontroller control, dual DC motors/ESCs, sensors, and remote input.
- Wired power distribution and common-ground connections across motors, controllers, sensors, and feedback subsystems.
- Programmed and debugged drive logic, throttle response, obstacle sensing, and button-triggered behaviors to improve reliability under load.

Automated Bridge Project - Academic Project

Apr 2026 - May 2026

- Built a motorized bridge prototype that opens/closes automatically using sensor input, traffic LEDs, and Arduino-based control logic.
- Integrated mechanical supports, bridge movement, and electronics to simulate safe boat/vehicle crossing behavior.

Door Alarm System - Personal Project

Jan 2026 - Feb 2026

- Built an ESP32-based door alarm with a reed switch, DFPlayer Mini, microSD card, and speaker to play custom audio alerts.
- Implemented UART communication and sensor-state logic to trigger MP3 playback when door/latch movement was detected.

Involvements

UNC Charlotte Robotics Fighting Club

Jul 2025 - Present

- Participated in robotics meetings and technical discussions related to robot design, mechanical systems, and electronics.